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Epistemological and methodological contributions of design research labs to transformation research

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Abstract

Amidst escalating ecological crises and societal challenges, exploratory spaces such as Real-World Laboratories (RWL) and Design Research Labs (DRL) have become central to researching transformation. While RWL are well-established as transdisciplinary spaces for co-creation and reflexive learning, DRL offer creativity, iterative testing and tangible prototyping. However, the relationship between RWL and DRL has received little systematic attention. This paper examines how DRL, situated predominantly within art and design universities, contribute to sustainability transformations. Building on a comparative cross-analysis of the defining characteristics of both frameworks, we investigate overlaps, divergences, and complementarities between RWL and DRL. Findings highlight that while both frameworks share participatory and experimental orientations, they foreground different yet complementary forms of knowledge production. Whereas RWL emphasise transdisciplinary collaboration and intervention in societal contexts, DRL foreground designerly epistemologies grounded in materialisation, prototyping, and speculative imagination. These practices enable alternative futures to be explored, materialised, and negotiated through tangible artefacts and interventions. The paper concludes by discussing opportunities for mutual learning and future collaboration between design research and transformation research.

Keywords Sustainability transitions, Transdisciplinary collaboration, Designerly epistemologies, Co-creation, Participatory design, Prototyping, Future-making, Boundary organisations, Intermediaries

Introduction: bridging design research and transformation research

In times of accelerating climate crises, social inequalities, and technocratic rationalities, calls for societal transformation have become ubiquitous. Yet, the environments in which transformative knowledge is produced remain largely tied to silo structures or traditional research formats (Fazey et al. 2020). Lately, the political and institutional landscape has increasingly adopted experimental spaces, such as innovation laboratories, hackathons, foresight exercises, real-world laboratories, policy labs, design research labs, and co-creation platforms as key instruments for transformation (Beekman et



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al. 2025; BMW 2025; Gast, Liedtke and Schmidt 2024; Graham and Milligan 2019; McCrory et al. 2022; Schöpke et al. 2017; Von Wirth et al. 2019; Voytenko et al. 2016). These *regulatory sandboxes* enable innovations to be tested temporarily under real-life conditions. Thus, they seek to connect knowledge production more directly to societal change by enabling situated experimentation, participation of multiple actors and collaborative learning and challenging established hierarchies of expertise (Peukert 2022). Among them, Real-World Laboratories (RWL) and Design Research Labs (DRL) are particularly relevant because both operate as spaces where knowledge production and societal transformation intersect.

Both RWL and DRL are experimental, participatory, reflexive, and future-oriented spaces that seek to address societal challenges through collaboration, iterative learning, and the integration of multiple forms of knowledge. While RWL have become a prominent approach within transformation research (Bason 2014; Parodi and Steglich 2021; Parodi et al. 2024; Richardson et al. 2026; Schöpke et al. 2017), DRL have received considerably less attention in this context despite sharing several core characteristics (Augsten and Harles 2024; Binder 2007; Binder and Brandt 2008; Binder et al. 2011; Boyer 2020; Cels, de Jong and Nauta 2012; Mouchrek and Krucken 2020; Reynolds-Cuellar et al. 2020). Despite these similarities, the relationship between the two approaches has received little systematic attention. This is surprising because both operate at the intersection of knowledge production and societal transformation while originating from different research traditions. Understanding their commonalities and complementarities is therefore important not only for strengthening collaboration between design research and transformation research but also for broadening the dialogue on epistemological and methodological approaches available for addressing complex sustainability challenges in RWL discourse.

Beyond its established contributions to sustainable production systems (Gast, Liedtke and Schmidt 2024; Liedtke et al. 2015b; Raatz et al. 2020; WBGU 2011) and participatory processes (Lähteenoja et al. 2023; Liedtke et al. 2015a, 2020; Matti and Rissola 2022; Sanders and Stappers 2014; Van Buuren, Lewis and Peters 2023), design becomes recognised as a driver of broader cultural transformation (Banz 2016; Ceschin and Gaziulusoy 2019; Collier and Gruendel 2022; Fineder et al. 2025; Jonas et al. 2016; Moussaoui, Krois and Palmieri 2025; Sommer and Welzer 2015). Design interventions as such are intentional and often experimental actions that seek not only to develop products or services but also to reframe problems, provoke reflection, and catalyse change in social, organisational, and ecological contexts (Dorst 2015; Banz 2016; Ceschin and Gaziulusoy 2019, Augsten 2022). Rather than treating cultural change as a by-product of technological or policy innovation, designers create interventions in public spaces through which new habits, values, and collective imaginaries can be explored, tested, and adopted (Collier and Gruendel 2022; Sommer and Welzer 2015). In this regard, design acts as a form of cultural production and world-making that mediates between systemic visions and lived realities (Cels, de Jong and Nauta 2012), highlighting its relevance for transformation-oriented research.

Characteristics such as co-creation, participation, exploratory and creative problem-solving, and a strong orientation toward the future are widely recognised as key features of research in RWL. At the same time, these qualities also constitute core competencies of design and form the basis of what design research conceptualises as a *designerly* way

of knowing (Cross 2024). In this article, *design research* is understood as a field of creative knowledge production. The term *designerly* is used to denote specific epistemological practices through which knowledge emerges through making, prototyping, iteration, visualisation and speculation (Cross 2024; Mareis 2011, 2012, 2022). *Design practices* such as prototyping, experimentation, co-design and speculative inquiry foreground material and imaginative engagement with *wicked problems* (Buchanan 1992; Rittel and Webber 1973). *Design interventions* can be understood as situated applications of these practices. They do not merely support transformation processes but constitute a distinct mode of inquiry through which alternative futures can be explored, materialised, and negotiated.

Thus, it is striking that, although the role of design has become established within international transformation-oriented initiatives (cf. Desis Network n.d., RISE n.d.), its epistemological and methodological contributions remain only partially reflected within the RWL discourse in Germany. Consequently, designers are still rarely positioned as research actors within RWL settings, while design is often reduced to a supportive role in facilitation or communication rather than recognised as a distinct mode of knowledge production. One possible explanation lies in a limited awareness within transformation research of the epistemological and methodological overlaps between the two fields. Although design methods and their iterative processes of materialisation are increasingly applied within RWL contexts, the forms of knowledge generated through material experimentation and prototyping remain comparatively underexamined. These observations motivated us, as design researchers, to comparatively analyse the two concepts of RWL and DRL.

In this paper, DRL are understood as experimental laboratories embedded in design and art universities in which designerly principles and practices are structurally anchored (Augsten and Harles 2024). Like RWL, DRL operate as “agents of change” (Cels, de Jong and Nauta 2012; Mouchrek and Krucken 2020) and contribute a particular form of knowledge concerning material experimentation, future-making, cultural critique, and world-making (Binder 2007; Binder and Brandt 2008; Binder et al. 2011; Boyer 2020; Reynolds-Cuellar et al. 2020). While RWL primarily pursue systemic transformation through interventions in real-world contexts, DRL often create protected spaces in which alternative futures can be explored through tangible artefacts, speculative scenarios, and experimental practices.

But rather than positioning one as superior to the other, we understand DRL and RWL as complementary experimental spaces with distinct strengths and limitations. We argue that examining their commonalities and complementarities offers an opportunity for mutual learning and thereby strengthening the dialogue between design research and transformation research. To explore this bridge, the paper conducts a comparative analysis of the two frameworks of DRL and RWL. The aim is not to privilege one framework over the other, but to identify areas of overlap, complementarity and potentials for mutual learning. In doing so, the paper seeks to clarify the contribution of designerly epistemologies to transformation research by strengthening the bridge between DRL and RWL.

The paper addresses the following research questions:

- How do the frameworks of DRL and RWL overlap and differ in their methodological and epistemological characteristics?

- How might designerly knowledge practices of DRL complement the methodological repertoire of RWL?
- What complementary contributions can DRL and RWL offer for transformation-oriented knowledge production?

Framework: design research labs and real-world laboratories as complementary experimental frameworks

Designerly fields of action in transformation research

Systems of production and consumption

Although design is increasingly recognised within transformative agendas, it is still most often framed as a supportive tool. Its contribution is primarily understood across three fields of action: First, systems of production and consumption, which must be redesigned to support ecological transition. These systems can range from circular economy initiatives to right-to-repair policies (cf. Gast, Liedtke and Schmidt 2024; Raatz et al. 2020). In this field, design practices are expected to play a central role in shaping the transition toward new forms of prosperity, consumption, and resource and energy usage (Liedtke et al. 2015b; Raatz et al. 2020; WBGU 2011).

Co-creation and participation

Second, design practices of co-creation and participation such as collaborative prototyping, scenario building, embodied mapping, or design thinking, which are essential for engaging communities in shaping transformation and negotiating conflicts (Lähteenoja et al. 2023; Liedtke et al. 2015a, 2020; Matti and Rissola 2022; Sanders and Stappers 2014; Van Buuren et al. 2023) are also highlighted by European and national research funding schemes (e.g., BMFTR; EU Innovation Policy). Both aim to involve affected communities in shaping sustainable solutions, encourage lifestyle changes, and support the balancing of conflicting goals. They explicitly refer to the transformative capacities of design (BMFTR 2023). However, co-creation and participatory approaches often reflect a narrow understanding of design, primarily focused on the development of sustainable products or services, while design's potential extends far beyond improving the energy and resource efficiency of consumer goods and includes shaping social practices, collective imaginaries, and transformative processes through designerly forms of inquiry and intervention.

Facilitation of a cultural shift

Third, designers and design practices occupy a recognised and deliberate role in shaping eco-social interventions that facilitate a cultural shift (Banz 2016; Ceschin and Gaziulusoy 2019; Collier and Gruendel 2022; Fineder et al. 2025; Jonas et al. 2016; Moussaoui, Krois and Palmieri 2025). Those interventions are intentional, often experimental actions introduced by designers to disrupt existing systems, practices, or behaviours and open up possibilities for change. Rather than producing only artefacts, they aim to reframe problems, provoke reflection, and catalyse transformation in social, organisational, or ecological contexts (Dorst 2015). Designers frequently use eco-social interventions in complex situations where linear solutions are insufficient and learning must occur through situated action. Transformation researchers such as Sommer and Welzer (2015) argue that design is not merely an ecological endeavour focused on “greener” products

but fundamentally enable a form of cultural production and reproduction (Sommer and Welzer 2015). Particularly, design interventions highlight design's unique ability to mediate between systemic visions and lived realities. Rather than treating cultural change as a by-product of technological or policy innovation, designers create interventions where new habits, values, and imaginaries can be explored, tested, and adopted. This positions design as a key agent in reshaping how societies relate to their environments, technologies, and collective futures (Cels, de Jong and Nauta 2012).

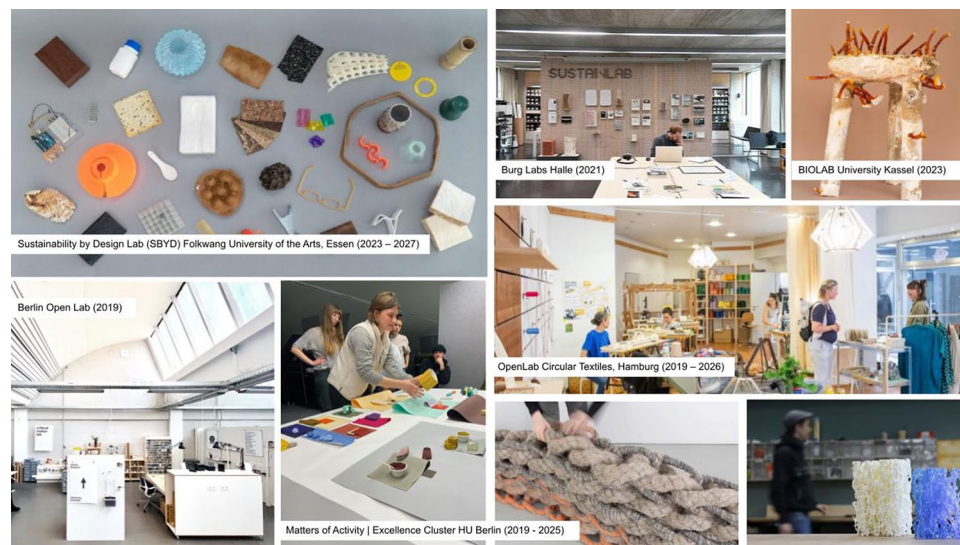
Design research labs and their transformative potential

Historical roots of design research labs

DRL have strong roots in design education and practice, from MIT's Media Lab (1985) and Stanford's Centre for Design Research (1984) to the Helsinki Design Lab (1968–2013) (Boyer 2020), which integrated strategic design into societal and governmental processes (Augsten and Harles 2024). Since the early 2000s, particularly in Europe, DRL have taken shape as research spaces within art and design universities, opening institutions to society through interdisciplinary research and third mission agendas (Augsten and Harles 2024; Binder and Brandt 2008; Dreyer, McTeague and Thoring 2025). In Germany, DRL foster co-creation, prototyping, and speculative world-making (Harles and Heidingsfelder 2021) while interrogating the competencies needed in transformative contexts (Peukert and Augsten 2025). Gast, Liedtke and Schmidt (2024) recognised DRL as crucial for eco-social innovation and characterised their approach as “bottom-up, practice-driven” evolution rather than centralised governance, fostering unique experimental spaces where theoretical frameworks meet tangible outcomes through material experimentation and interdisciplinary collaboration. Their practices merge academic inquiry and practice-based experimentation, positioning DRL simultaneously as experimental spaces for societal transformation and as drivers redefining the role of design in transdisciplinary research, transformation, and public engagement. Furthermore, DRL simultaneously reflect on design education and necessary competencies for designers within transformative research contexts (Ibach et al. 2025). In this article, DRL are understood as institutionalised spaces of creative and open experimentation, where research and education converge to actively engage with transformative agendas structurally embedded within art and design universities, operating as transdisciplinary and practice-oriented spaces.

Selected cases of design research labs

We selected six examples of DRL, which differ in their thematic focus and geographical distribution, as well as their approaches to knowledge production (see Visual 1). They address a range of transformation challenges such as new cultures of materials (examples 5, 6, and 2), social innovation at the nexus of technology, society, and the arts (4), tangible sustainability interventions (1), and ecological or more-than-human design (1, 3). At the same time, their institutionalisation reflects diverse funding and organisational models. Some operate as time-bound, publicly funded research projects aimed at establishing new spaces and transfer mechanisms (1, 6), while others function as long-term institutional platforms embedded within art and design universities and supported through a combination of institutional funding and third-party research collaborations (5). Across all six cases, the institutional diversity highlights the role of DRL as evolving



Visual 1 Table of DRL mentioned in 2.2.2: (1) Sustainability by Design Lab (SBYD) Folkwang University of the Arts, Essen; (2) BurgLabs, Burg Giebichenstein University of Arts and Design Halle; (3) BIOLAB Kassel University; Berlin; (4) Open Lab, University of the Arts Berlin; (5) Matters of Activity – Excellence Cluster, Humboldt University Berlin; and (6) Open Lab Circular Textiles, Hamburg (Berlin Open Lab: [About](#); Berlin Open Lab: [Co-Working Space](#); Biolab Kassel; Cluster of Excellence. "Matters of Activity. Image Space Material". [Design Lab](#); Cluster of Excellence. "Matters of Activity. Image Space Material". [MoA Design Research Lab](#); Cluster of Excellence. "Matters of Activity. Image Space Material". [MoA Design Research Lab](#); OpenLab Circular Textiles; [Rausgegangen](#))

research environments that may contribute to transformation research through designerly knowledge production.

Building on four key characteristics of DRL (Augsten and Harles 2024), this article inquires how the epistemological and methodological approaches of DRL entangle with those of RWL to explore potentials for mutual learning. We argue that DRL offer complementary contributions to RWL not merely through creative methods such as prototyping, co-creation, and visualisation, but by introducing a distinct designerly knowledge practice that complement existing approaches in RWL and might forge stronger alliances between design research and transformation research.

Four key characteristics of design research labs

The following key characteristics are derived from prior conceptualisations of DRL in design research literature (Augsten and Harles 2024) based on a case analysis and from our own analytical perspective as design researchers engaged in lab-based research. They do not serve to distinguish DRL from other experimental spaces, such as RWL or Living Labs, which share important features including transdisciplinarity, experimentation, and societal engagement. Rather, these characteristics are used as analytical dimensions to articulate how DRL organise knowledge production through designerly epistemological and methodological practices. From a design research perspective, particular emphasis lies in practices such as materialisation, prototyping, speculation, and the involvement of actors such as designers, craftspeople, and artists, whose methods and forms of knowledge production differ from those foregrounded in transformation research. While some of these key characteristics conceptually overlap with other lab formats, DRL exemplify them through distinct designerly approaches.

The four key characteristics presented provide a conceptual framework for understanding DRLs' epistemological and methodological stances and serve as a basis for the subsequent comparative framework analysis of DRL and RWL (Section "[Comparative cross-analysis of the frameworks: epistemological and methodological proximity of design research labs and real-world laboratories](#)"). Synthesising these characteristics reveals how knowledge production in DRL exemplifies a designerly epistemology (Cross 2024; Mareis 2011, 2012) that foregrounds material and imaginative engagement with wicked problems and reconfigures how knowledge is generated, negotiated, and applied in processes of societal transformation. While questions of participation, inclusion, and power relations are thematically central to many DRL projects, they are not structurally predefined by the format itself. Rather, they are actively negotiated through designerly practices, in which designers facilitate collaboration and create material and methodological conditions for diverse actors and forms of knowledge to engage in designerly research processes.

- 1 **Fostering real-life innovations:** DRL bridge the gap between art and design universities and real-world contexts by situating design practices directly within everyday life (Binder and Brandt 2008). DRL function as platforms for collaborative inquiry, where designers, affected communities, and strategic partners actively participate in experimentation. Knowledge is co-produced through prototyping and testing and situated where interventions are applied. Such immersion ensures that innovations are epistemically grounded in lived experience rather than based on abstract problem definitions, thereby embedding practice-based learning within processes of transformation.
- 2 **Acting responsible and mission-oriented:** DRL align their research agendas towards pressing societal challenges rather than on theoretical aspects of design. Their mission orientation resonates with transformative research frameworks such as Responsible Research and Innovation, which emphasise societal relevance, ethical responsibility, and the engagement of multiple actors (Timmermans et al. 2020; Von Schomberg 2013). Projects situated within DRL are guided by questions of social and environmental sustainability. This can involve sustainable practices, renewable energies and materials, or the inclusion of marginalised voices. DRL move beyond academic insularity to ensure that design interventions serve broader societal needs.
- 3 **Crafting collaborative images of the future:** Operating under conditions of uncertainty and complexity, DRL create exploratory atmospheres for anticipating and shaping possible futures. They employ speculative and critical design practices such as design fiction, speculative enactments, and tangible "as-if" prototypes (Dunne and Raby 2013; Elsdén et al. 2017) to engage diverse actors in envisioning the long-term societal implications of emerging technologies and ecological shifts. This is called *future-making*. These tools and methods translate complex scenarios into experiential, tangible, and visible forms and foster dialogue and negotiation among participants.
- 4 **Promoting a multiplicity of knowledges:** Although DRL may appear as experimental workspaces, they expand knowledge production beyond academic boundaries and contribute to a paradigm shift toward socially robust knowledge (Carayannis, Campbell and Rehman 2016; Nowotny, Scott and Gibbons 2003). Depending on thematic focus, they aim to integrate diverse forms of knowledge such as local or situated knowledge of non-academics as everyday experts, craftspeople, Indigenous communities, or

artists. This epistemic openness is reflected in their methodological diversity, which draws from design, craft, social sciences, and natural sciences. Furthermore, DRL unite varied modes of knowledge transfer, including prototypes, exhibitions, seminars, and workshops.

Real-world laboratories as experimental framework for transformative knowledge production

Real-world laboratories as transformative research settings

RWL are spaces of transdisciplinary sustainability research. In German-speaking countries, this approach has become widely established since the early 2010s (Schäpke et al. 2018a, 2018b, 2018c). Defined by the WBGU (2016) as “scientifically designed spaces of collaborative sustainability research involving intervention,” they generate system, target, and transformation knowledge (ProClim 1997) through on-site co-creative interventions, integrating academic and local expertise (Schneidewind and Singer-Brodowski 2015) and facilitating iterative learning via real-world experiments. They aim to initiate and scientifically support sustainability transitions through participatory experimentation in real-world contexts (Parodi, Steglich and Bylund 2023; Schäpke et al. 2024). While rooted in the legacy of transdisciplinary research, RWL extend it through their explicitly action- and impact-oriented approach (Beecroft and Parodi 2016; Defila and Di Giulio 2018). Their epistemological conception builds on the co-production of contextualised, socially robust knowledge and its reflexive assessment (Schäpke et al. 2024), supported by transdisciplinary collaboration, long-term monitoring, and methodological reflection (Beecroft and Parodi 2016).

This orientation towards societal impact reflects a broader paradigm shift in research: traditional approaches aimed at generating universal, generalisable knowledge often fall short when addressing complex, context-dependent problems. In response, science has opened up to transformative approaches that combine disciplinary with practice-based, experiential, and local knowledge. This shift reflects a fundamental reconceptualisation of the laboratory itself: no longer a secluded, controlled environment, but a situated, permeable space for collaborative inquiry and societal intervention. In this sense, RWL exemplify how research is adapting its epistemological and methodological foundations to meet the demands of transformation-oriented science, where generating socially robust knowledge and contributing directly to change processes are equally central (Wanner et al. 2018).

Five key characteristics of real-world laboratories

Based on current literature (Augenstein et al. 2022, 2026; Bernert et al. 2024; Christ et al. 2024; Erisman et al. 2024; Kreß-Ludwig et al. 2024; Schäpke et al. 2024; Wanner et al. 2024), we distilled five key characteristics that distinguish RWL as a research approach, structure how they operate, and highlight their specific contribution to transformative research.

- 1 Integrating transdisciplinary perspectives:** RWL are characterised by the systematic integration of actors from science, business and industry, politics, administration, and civil society. This integration not only follows the goal of integrating different bodies of knowledge but also the goal of social learning processes, collective decision-making, and joint problem-solving (Bernert et al. 2024; Schäpke et al. 2024).

- 2 **Experimenting in real-world contexts:** In RWL, working on pressing challenges takes place in real-world environments (e.g., urban neighbourhoods, energy systems, mobility infrastructures). The focus is on experimental, reversible actions that address socially relevant issues which are dealt with in open processes (Christ et al. 2024; Kreß-Ludwig et al. 2024).
- 3 **Aspiring epistemological openness:** Knowledge production in RWL is characterised by reflexivity towards normative assumptions, power relations, and conflicting goals (Erisman et al. 2024). RWL recognise that knowledge is not neutral but arises in a context-bound, perspectival, and relational manner (Augenstein et al. 2022). New epistemological approaches are co-produced through a diversity of actors.
- 4 **Pursuing processuality and iteration:** RWL operate in a processual manner, with iterative cycles of problem understanding, intervention, observation, and reflection, and are systematically accompanied by research. This openness is necessary to deal with the inherent uncertainty and dynamics of transformative processes (Wanner et al. 2024).
- 5 **Orienting research toward impact:** A key objective of RWL is to generate impact, understood as social, political, ecological, or cultural change. The impact dimension encompasses (1) individual and collective learning processes, such as the development of sufficiency practices, shifts in values, and new forms of everyday action; (2) new forms of governance and cooperation networks, including participatory circular societies and novel constellations of actor collaboration; and (3) physical and institutional changes in space, such as the transformation of infrastructures, regulations, and organisational arrangements (Bernert et al. 2024; Kreß-Ludwig et al. 2024; Schäpke et al. 2024).

Comparative cross-analysis of the frameworks: epistemological and methodological proximity of design research labs and real-world laboratories

Methodological rationale

Unlike comparative case study approaches that analyse empirical cases (cf. Blair-Walcott 2023; Heath 2024), this article compares two conceptual frameworks of DRL and RWL which were derived from the literature (cf. Section “[Framework: design research labs and real-world laboratories as complementary experimental frameworks](#)”). Even though both frameworks and their defining key characteristics originate from previous empirical work, the analysis does not compare empirical manifestations of DRL and RWL, nor does it seek to construct ideal types. Instead, it compares conceptual frameworks by treating their defining characteristics as analytical dimensions of knowledge production. As Table 1 shows, both frameworks emerged from different disciplinary traditions and were developed using different analytical categories.

Table 1 Overview of categories of RWL and DRL

Categories	DRL	RWL
Epistemological orientation	Designerly epistemologies	Transdisciplinary sustainability research
Institutional embedding	Art and design universities	Sustainability research institutions, universities, municipalities
Actors involved	Designers, artists, craftspeople, researchers	Researchers, citizens, policy-makers, practitioners
Methods employed	Prototyping, speculative design, visualisation, making	Co-creation, experimentation, intervention, evaluation

Our interest lies in the qualities of creative knowledge production that characterise both DRL and RWL. Although these qualities often become visible through specific methods, they are analytically situated at a higher level of abstraction and capture broader epistemological orientations rather than methodological choices alone. We therefore applied a comparative framework cross-analysis, treating the key characteristics of each lab type as analytical dimensions rather than fixed variables. These key characteristics (cf. Section “[Four key characteristics of design research labs](#)” and “[Framework: design research labs and real-world laboratories as complementary experimental frameworks](#)”) are understood as qualities of creative knowledge production such as openness, multiplicity of knowledge, or crafting that shape how knowledge is generated, negotiated, and validated. This perspective enables a reciprocal interpretation of the two frameworks, allowing the key characteristics of one laboratory to illuminate and contextualise those of the other.

In a first step, the four key characteristics of DRL (cf. “[Four key characteristics of design research labs](#)”) are used as analytical dimensions through which the framework of RWL is examined (Table 2). In a second step, the five key characteristics of RWL (cf. “[Five key characteristics of real-world laboratories](#)”) are applied to analyse the framework of DRL (Table 3). These key characteristics are not methodological tools in themselves but represent underlying qualities of knowledge production. They capture epistemic orientations such as multiplicity of knowledge, crafting, and openness that inform and shape research practices. Although these qualities may become visible through specific methods, they extend beyond methodological choices and reflect broader ways of producing, relating, and validating knowledge. In both Tables 2 and 3, the description *limited, partial and strong* for mutual learning in the third column of the tables indicate the extent to which a characteristic appears absent, partially present, or underdeveloped in the respective framework.

This approach allows us to identify areas of overlap, complementarity, divergence, and indicates potential for mutual learning between the two frameworks in order to explore how distinct epistemological and methodological orientations become visible when viewed through the analytical dimensions of the respective other framework. Nevertheless, it should be acknowledged that this qualitative analysis might be perceived as

Table 2 Analysis of RWL by DRL key characteristics

DRL key characteristic	Application to RWL	Result of analysis
Fostering real-life innovations	RWL explicitly foster real-world innovations through interventions, prototypes, and experiments directly in socio-ecological contexts.	Limited potential for mutual learning: RWL are by definition action-oriented, producing practical solutions with societal impact.
Acting responsible and mission-oriented	RWL often have explicit sustainability or transformation missions, addressing societal and environmental challenges responsibly.	Limited potential for mutual learning: RWL combine responsible, reflexive, and long-term system thinking, often beyond project outputs.
Crafting collaborative images of the future	RWL co-create visions with diverse actors, including citizens, policymakers, and organisations, often exploring desired futures.	Partial potential for mutual learning: The participatory and systemic orientation of RWL supports collaborative future-making, although they focus more on socio-ecological outcomes than design artefacts.
Promoting a multiplicity of knowledge	RWL integrate scientific, local, tacit, and practitioner knowledge, fostering transdisciplinary learning.	Limited potential for mutual learning: Knowledge plurality is central; RWL value co-production of knowledge across disciplines and sectors.

Table 3 Analysis of DRL by RWL key characteristics

RWL Key Characteristics	Application to DRL	Result of analysis
Integrating transdisciplinary perspectives	DRL involve multiple perspectives (designers, people from various scientific disciplines, users, and sometimes external partners), but only occasionally integrate the full breadth of trans-disciplinary actors as RWL do.	Strong potential for mutual learning: Engagement is usually bound to the design project or learning context; DRL are multi- and transdisciplinary (meaning including bodies of knowledge beyond academia), and cross-sectoral involvement is less pronounced and dependent on the project focus than in RWL.
Experimenting in real-world contexts	DRL implement tangible and visible prototypes and interventions, sometimes in real-world contexts, but often in bounded project or educational settings inside design universities.	Strong potential for mutual learning: Interventions have a strong real-world connection but are typically smaller-scale, controlled, or pedagogically framed, and less embedded in complex socio-ecological systems.
Aspiring epistemological openness	DRL aim to develop transformative competencies, innovative solutions, and new design knowledge.	Partial potential for mutual learning: Epistemological openness in DRL refers more to experimental future-making and is more related to a framework of design epistemology than transdisciplinary epistemologies.
Pursuing processuality and iteration	Iteration is a core strength of DRL: cycles of prototyping, testing, and reflection are central to the design process.	Limited potential for mutual learning: DRL excel in iterative, process-oriented approaches as core to a designerly approach.
Orienting research toward impact	DRL generate impact primarily in the domains of knowledge production, capacity building, and material innovation. Their effects are often visible in academic outputs, knowledge transfer formats such as exhibitions or installations, new methods, or the learning of participants, but less in terms of established measurable systemic transformation.	Partial potential for mutual learning: Impact is noticeable but bounded and smaller in scope compared to the long-term, systemic impact ambitions of RWL related more closely to policy-making.

interpretive. The comparison of the key characteristics of the two frameworks is reconstructed from the literature, which was based on empirical data. However, the results of the analysis are qualitative, based on a collaborative and discursive process by the authors. Consequently, the findings should be understood as a heuristic contribution that identifies potential areas of overlap and for mutual learning to understand the contribution of design to transformation research better.

Applying DRL characteristics as analytical dimensions to the RWL framework

RWL naturally resonate with many of the DRL characteristics, even though these principles were originally formulated for design contexts (Table 2). *Fostering real-life innovations* is a core feature of RWL: experiments and interventions are implemented in real socio-technical and ecological contexts, creating actionable knowledge and tangible solutions. *Acting responsible and in a mission-oriented manner* is also central: RWL explicitly aim to contribute to sustainability and systemic transformation, guided by ethical and societal commitments. In terms of *crafting collaborative images of the future*, RWL actively involve multiple actors in envisioning desirable futures, from local communities to policymakers, aligning well with DRL's emphasis on co-created future scenarios. Due to different methodological traditions, co-creative methods range from more science-based approaches to craft-based and tangible forms of inquiry which inform the different ways of creative knowledge production.

Finally, promoting a *multiplicity of knowledge* is embedded in the transdisciplinary approach of RWL as well, integrating scientific insights, practitioner know-how, and

local or tacit knowledge to tackle complex real-world problems. Altogether, RWL show only limited potential for mutual learning for DRL principles 1,2 and 4, but their transdisciplinary implementation is often broader in scale, more embedded in real-world systems, and oriented toward societal transformation rather than design artefacts. But the analysis indicates a partial potential for mutual learning concerning the characteristic *Crafting collaborative images of the future*. This could be, for instance, the prototyping of tangible artefacts to enhance collaborative and creative knowledge production in RWL research.

Applying RWL characteristics as analytical dimensions to the DRL framework

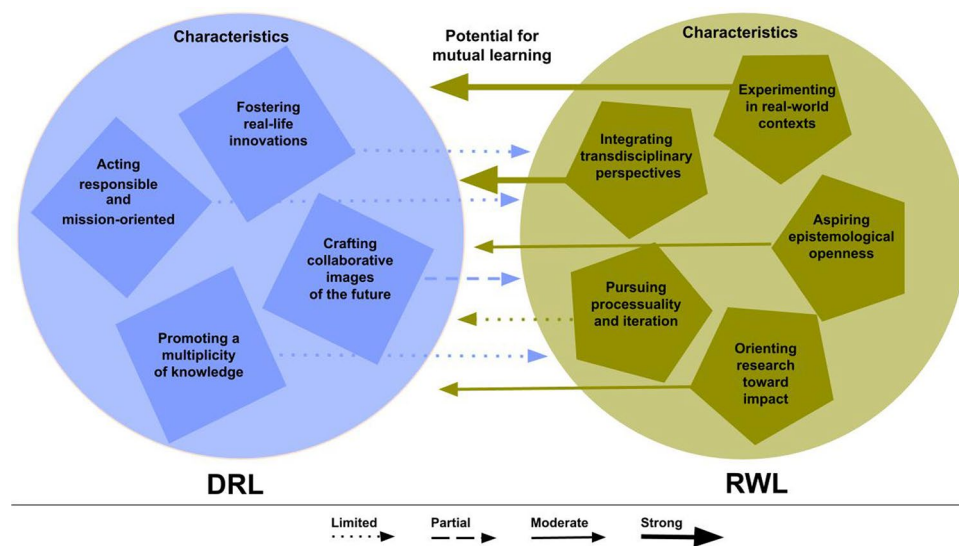
In a second analytical step, DRL were examined through the framework of RWL characteristics, revealing fields for mutual learning, additional overlaps and more pronounced distinctions (Table 3). DRL do incorporate multiple actors, but usually within bounded project or educational contexts rather than on entire socio-technical systems. Similarly, real-world interventions exist in DRL through prototypes, workshops, and pilot projects, yet these are typically controlled or pedagogically framed, rather than embedded in complex socio-ecological systems as in RWL. In addition, DRL contribute to systemic transformation in a more limited sense: they develop transformative competencies, design interventions, or innovative practices, but their impact is primarily at the level of learning or artefacts, less on entire socio-technical systems. Reflexivity is present in DRL, focused on improving design processes and outcomes, but the consideration of wider societal or ecological consequences is less explicit, which indicates another area for mutual learning.

Discussion: complementary knowledge practices in design research and transformation research

The comparative framework analysis revealed substantial overlap between DRL and RWL, particularly regarding their commitment to experimentation, participation, societal relevance, and the integration of diverse forms of knowledge. At the same time, the analysis identified distinct epistemological orientations, methodological emphases, and institutional settings that shape how knowledge is produced and applied within each framework. Rather than positioning one format as superior to the other, the findings suggest a complementary relationship in which both contribute different yet mutually enriching approaches to transformation-oriented research. Visual 2 summarises these areas of potential mutual learning. Building on our findings, the following discussion examines the complementary epistemological and methodological contributions of DRL and RWL and reflects on design research's implications for transformation research.

Epistemological dimension

Epistemologically, RWL and DRL operate from distinct yet complementary positions. RWL are grounded in transdisciplinary sustainability research, with a strong orientation towards contextuality, reflexivity, and societal robustness (Augenstein et al. 2022; Schöpke et al. 2024). In this sense, they represent a *Mode 2* understanding of science (Nowotny, Scott and Gibbons 2003) in which normative approaches, social relevance, scientific quality, and political effectiveness are intertwined. RWL address systemic, non-linear challenges and actively embrace ambivalence and uncertainty, as productive



Visual 2 Comparison of DRL and RWL characteristics and their potential for mutual methodological and epistemological learning

components of transformation processes (Schneidewind et al. 2018; Wagner and Grunwald 2019). Their epistemic focus lies in integrating diverse forms of knowledge to generate socially robust as well as policy-relevant knowledge capable of informing systemic change and governance in sustainability transformations.

By contrast, insights in DRL primarily emerge through designerly processes of inquiry. DRL embody a designerly epistemology (Mareis 2011), in which knowledge emerges through materialisation, prototyping, and speculative imagination (Dunne and Raby 2013; Peukert and Vilsmaier 2021). Here, knowledge is not only co-produced among actors but materially and imaginatively enacted. The outputs of these processes serve as “epistemic things” (Rheinberger 1997) that allow possible futures to be explored, experienced, and negotiated before they become stabilised within institutional or governance frameworks. Consequently, research in DRL relies less on standardised scientific procedures and more on the situational development of context-specific repertoires of inquiry, often oriented toward the crafting of tangible outcomes or the facilitation of transformative learning processes. This epistemological stance positions DRL as spaces where design itself becomes a way of knowing, foregrounding imagination and material experimentation as important practices for societal transformation.

This difference indicates an area of complementarity rather than epistemological superiority. While RWL emphasise governance integration and the co-production of socially relevant knowledge, DRL foreground material, speculative, and future-oriented forms of inquiry. These orientations reflect different but complementary ways of producing knowledge for transformation. Rather than expanding the epistemological scope of transformation research in general, DRL contribute additional forms of knowledge production that remain comparatively less visible within established RWL frameworks.

Methodological dimension

A core feature of RWL is its methodological pluralism. Research in RWL includes qualitative, quantitative, participatory, and reflexive approaches, as well as designerly methods such as co-design, prototyping, and speculative approaches (Franck et al. 2024). New

evaluation approaches such as ‘formative evaluation’ (Wiefek, Nagy and Schäfer 2024) or ‘leverage points’ (Schäpke et al. 2024) aim to systematically assess impact and process quality (Ibach 2026). Thus, methods in RWL primarily function to structure transdisciplinary collaboration, guide intervention processes, and evaluate transformation outcomes over time.

DRL and RWL aim to shape transformative processes actively yet differ in their respective emphasis and in how methods are positioned in the research process. Although both are characterised by participation and openness, they diverge in their focus and practices. While RWL place participatory approaches, particularly collaboration with societal actors, often including marginalised groups, at the forefront, DRL focus more strongly on the concept of *future-making*, which is the design-oriented development of sustainable futures (Ehn, Nilsson and Topgaard 2014; Fry and Nocek 2021; Hunt 2017; Mazé 2016; Rawsthorn 2018; Smith, Baille and McHattie 2017; Yelavich and Adams 2014). In DRL, design operates as a specific epistemic practice that makes possible futures tangible and negotiable (Ibach 2024; 2025). Design methods therefore act not only as tools but as modes of inquiry through which research questions are reframed and multiple futures explored.

In DRL, responsibility is primarily expressed at the level of researchers’ attitude and ecological concerns, whereas participatory or interdisciplinary elements vary depending on the institutional context (e.g., interdisciplinary at MoA (5) or participatory at SBYD(1), see Visual 1). By contrast, within RWL the social dimension of sustainability and the active involvement of diverse societal groups are more structurally embedded in the research approach. However, the focus of RWL remains primarily human-centred, emphasising the participation and coordination of multiple actors. In contrast, DRL more frequently engage with more-than-human or multispecies design, explicitly addressing the needs, agencies, and coexistence of multiple species. Combining both frameworks may allow transformation research to connect early-stage imagination and material experimentation with later-stage implementation, governance embedding, and systemic learning.

Complementary contributions of design research lab and real-world laboratories to transformation research

The differences outlined above highlight the complementary roles of RWL and DRL: RWL provide the institutional and systemic frameworks for societal transformation, while DRL contribute the imaginative, material, and aesthetic dimensions that open up new perspectives and imaginaries for change. In this regard, DRL can function as what Moussaoui, Krois and Palmieri (2025) call “catalysts of transformation”. Their research activities take place across all three transformative fields of action (see Section. “[Designerly fields of action in transformation research](#)”): in systems of production and consumption, in processes of co-creation and participation, and in transformations of cultural imaginaries. Whereas RWL aim at systemic transformation through real-world interventions, DRL operate mainly in environments of art and design universities, where transformation is primarily directed towards learning and experimenting.

Our analysis suggests that the strongest overlaps (Visual 2) indicate a substantial common ground between DRL and RWL. At the same time, areas identified as having partial potential for mutual learning reveal distinct knowledge practices and research

cultures. Rather than representing deficiencies, these differences point to complementary contributions that can broaden the repertoire of transformation-oriented research. Furthermore, the cross-analysis indicates substantial overlap between both frameworks. Many characteristics associated with RWL can also be found in DRL settings, while DRL simultaneously encompass additional designerly practices such as speculative inquiry, material prototyping, and aesthetic exploration. This suggests that certain RWL interventions may be realised within DRL settings, while other DRL activities extend beyond the scope and objectives typically associated with RWL.

At the same time, DRL contribute complementary forms of creative knowledge production to transformation research. Whereas the central contribution of RWL as experimental spaces lies in fostering systemic shifts, DRL pursue multiple focal points such as relational knowledge production (Sametinger, Quadflieg and Augsten 2025; see examples 2, 3, 5), capacity building (see examples 1, 4, 6), and material innovation (see examples 2,3,5). These activities may include material development but can also contribute to broader societal transformation. This difference is partly rooted in their organisational structure. While RWL often emerge as project-based initiatives focused on specific transformation challenges, DRL are typically established as longer-term institutional structures that host multiple research projects under a shared designerly orientation. However, both display considerable organisational diversity, and examples of long-term institutionalised RWL can also be found.

In this sense, DRL function as boundary organisations (Parker and Crona 2012), operating multidirectionally and, in line with transformation research literature, as *intermediaries* that mediate between science, politics, and society (Kivimaa et al. 2019; Moss 2009). Their institutional embedding within art and design universities enables them not only to connect research with societal actors but also to integrate research, teaching, and curriculum development through student involvement and educational programmes. Thus, research projects within a DRL framework may pursue systemic shifts (Gaziulusoy and Brezet 2015) while contributing to creative knowledge production, capacity building, and education simultaneously. By contrast, although RWL may involve students and contribute to educational activities, such engagement is not structurally required and depends on the specific institutional and project context. As a result, the integration of teaching, student participation, and research is more systematically institutionalised in DRL than in most RWL settings. Thus, knowledge transfer between research and teaching is structurally embedded in the institutional anchoring of DRL.

Finally, DRL and RWL represent different but complementary environments for transformation-oriented research. Whereas RWL primarily emphasise the integration of transdisciplinary knowledge, DRL contribute additional designerly epistemologies and knowledge formats, such as more-than-human approaches and critical or speculative design, which are less prominent within established RWL frameworks (Dunne and Raby 2013; Harles 2024; Mareis 2011; Ståhl et al. 2023; Wilkie and Fariás 2015; Wilkie, Savransky and Rosengarten 2019). This distinction highlights the complementary relationship between both frameworks and underscores the potential of DRL to contribute additional epistemological and methodological perspectives to transformation research.

Conclusion: towards stronger synergies between design research and transformation research

This paper aims to strengthen the dialogue between design research and transformation research by clarifying how DRL and RWL differ both in their methodologies but also in their knowledge practices and institutional anchoring – and thus can learn from each other. Rather than viewing one framework as a subset of the other, the findings point towards opportunities for collaboration that combine the strengths of both approaches. The decisive distinction lies less in specific methods than in the breadth of knowledge practices supported by each format. While both DRL and RWL engage in experimentation, participation, and iterative learning, DRL additionally foreground designerly forms of knowledge production such as prototyping, material exploration, and speculative inquiry. These practices complement the strengths of RWL and may broaden the repertoire of approaches available for transformation research.

From a policy perspective, three directions emerge. First, funding frameworks should recognise the dual nature of DRL as both long-term environments and spaces for design interventions. Stable institutional support combined with flexible third-party grants would enable DRL to sustain facilities while remaining adaptive. Second, policies should encourage cross-institutional collaboration between DRL and RWL to connect the imaginative and materialised capacities of design with the systemic, governance-oriented orientation of transformation research. Third, outreach and participation strategies need to lower entry barriers for diverse stakeholders, ensuring inclusivity beyond academic and artistic communities. Such measures would enhance the societal relevance of DRL and consolidate their role as boundary organisations that mediate between science, policy, and society.

Future research should therefore examine more closely how designerly epistemologies can contribute to and interact with transformation-oriented research, particularly through prototyping, material experimentation, and speculative approaches as complementary modes of knowledge production. This includes investigating how designerly practices can function as epistemological extensions of RWL, enabling earlier-stage exploration of transformation processes, including the articulation, negotiation, and materialisation of alternative futures. Further research should also explore how institutional collaborations between DRL and RWL can be organised to connect exploratory design inquiry with intervention-oriented transformation research, and how designerly knowledge practices can contribute to expanding the methodological repertoire of transformation research. By addressing these questions, future work can help to consolidate the role of DRL within transformation research and clarify their contribution to knowledge production for systemic societal change.

Abbreviations

DRL Design Research Labs
RWL Real-World Laboratories

Author contributions

All authors were actively and continuously engaged throughout the entire research and writing process. The paper was developed collaboratively using shared documents, with transparent versioning and regular meetings to ensure collective decision-making and critical reflection at each stage. Together, the authors co-conceptualized the idea of the paper and argumentation, designed the methodology, conducted the formal analysis, jointly drafted the manuscript, and designed tables and graphics. All authors contributed to reviewing, editing, and refining the final text. Andrea Augsten has led the overall coordination.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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